

Jihyeok Park (박지혁)

ASSISTANT PROFESSOR

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Research Interests

The ultimate goal is to help developers design and implement high-quality software using diverse programming language techniques:

- **program analysis** for automatically understanding program behaviors and detecting software bugs and vulnerabilities.
- **mechanized specification** to fill the gap between human-readable specifications and machine-friendly software.
- **program synthesis** to lessen the burden of software development by automatically generating programs.
- **automated testing** to generate test cases for software automatically on behalf of humans.

Education

Integrated M.S. & Ph.D in School of Computing

Daejeon, South Korea

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)

Mar. 2016 - Feb. 2022

- Ph.D. Thesis - JavaScript Static Analysis for Evolving Language Specifications
- Programming Language Research Group
- Advisor: Sukeyoung Ryu

B.S in School of Computing and Mathematical Science

Daejeon, South Korea

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)

Mar. 2012 - Feb. 2016

Work Experience

Mar. 2023 - PRESENT **Assistant Professor**, Korea University

Seoul, South Korea

Feb. 2022 - Feb. 2023 **Post Doctoral Fellow**, Oracle Labs

Brisbane, Australia

Publications

[SAS 2026] Synthesizing Parsing Expression Grammars via Analysis-based Pruning

Seongmin Ko and **Jihyeok Park**[†]

Static Analysis Symposium

[TOSEM 2026] Selective Feature-Sensitive Coverage for Conformance Testing of Programming Languages

Kanguk Lee, Seunghwan Kim, **Jihyeok Park**[†], and Sukeyoung Ryu[†]

Transactions on Software Engineering and Methodology

[ASE 2025] Debun: Detecting Bundled JavaScript Libraries on Web using Property-Order Graphs

Seojin Kim*, Sungmin Park*, and **Jihyeok Park**[†] (* equally contributed)

Proceedings of the IEEE/ACM International Conference on Automated Software Engineering

[FSE 2025 Demo] JSSpecVis: A JavaScript Language Specification Visualization Tool

Minseok Choe*, Kyungho Song*, Hyunjoon Kim, and **Jihyeok Park**[†] (* equally contributed)

Proceedings of the ACM International Conference on the Foundations of Software Engineering Demonstration track

[PLDI 2024] PL4XGL: A Programming Language Approach to Explainable Graph Learning

Minseok Jeon, **Jihyeok Park**[†], and Hakjoo Oh[†]

Proceedings of the 45th ACM SIGPLAN Conference on Programming Language Design and Implementation

[CACM 2024] JavaScript Language Design and Implementation in Tandem

Sukeyoung Ryu and **Jihyeok Park**

Communications of the ACM

[PLDI 2023] Feature-Sensitive Coverage for Conformance Testing of Programming Language Implementations

Jihyeok Park, Dongjun Youn, Kanguk Lee, and Sukeyoung Ryu

Proceedings of the 44th ACM SIGPLAN Conference on Programming Language Design and Implementation

[ESEC/FSE 2022] Automatically Deriving JavaScript Static Analyzers from Specifications using Meta-Level Static Analysis

Jihyeok Park, Seungmin An, and Sukyoung Ryu

Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

[PLDI 2022 Tutorial] Filling the Gap between the JavaScript Language Specification and Tools using the JISET Family

Sukyoung Ryu, Jihyeok Park, and Seungmin An

Proceedings of the 43rd ACM SIGPLAN Conference on Programming Language Design and Implementation Tutorial

[ASE 2021] JSTAR: JavaScript Specification Type Analyzer using Refinement

Jihyeok Park, Seungmin An, Wonho Shin, Yuseung Sim, and Sukyoung Ryu

Proceedings of the 36th IEEE/ACM International Conference on Automated Software Engineering

[ESEC/FSE 2021] Accelerating JavaScript Static Analysis via Dynamic Shortcuts

Joonyoung Park*, Jihyeok Park*, Dongjun Youn, and Sukyoung Ryu (* equally contributed)

Proceedings of the 29th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

[ECOOP/ISSTA 2021 DS] JavaScript Static Analysis with Evolving Engines and Specification

Jihyeok Park

Proceedings of the 35th European Conference on Object-Oriented Programming and Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis Doctoral Symposium track

[CSUR 2021] A Survey of Parametric Static Analysis

Jihyeok Park*, Hongki Lee*, and Sukyoung Ryu (* equally contributed)

ACM Computing Surveys, Volume 54, Issue 7, Article No. 149, pp 1–37

[ICSE 2021] JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification

Jihyeok Park, Seungmin An, Dongjun Youn, Gyeongwon Kim, and Sukyoung Ryu

Proceedings of the 43rd ACM/IEEE International Conference on Software Engineering

ACM SIGSOFT Distinguished Paper Award

[ASE 2020] JISET: JavaScript IR-based Semantics Extraction Toolchain

Jihyeok Park, Jihee Park, Seungmin An, and Sukyoung Ryu

Proceedings of the 35th IEEE/ACM International Conference on Automated Software Engineering

[IEEE Software 2019] Towards Analysis and Bug Finding of JavaScript Web Applications in the Wild

Sukyoung Ryu, Jihyeok Park, and Joonyoung Park

IEEE Software, Volume 36, Issue 3, pp 74-82

[Scala 2018] Path Dependent Types with Path-Equality

Jaemin Hong, Jihyeok Park, and Sukyoung Ryu

Proceedings of the 9th ACM SIGPLAN Symposium on Scala

[ProWeb 2018] A Framework for Dynamic Inter-Device Task Dispatch with Eventual Consistency

Jihyeok Park, Joonyoung Park, Yoonkyong Lee, Chul-Joo Kim, Byoungoh Kim, and Sukyoung Ryu

Proceedings of the 2nd International Workshop on Programming Technology for the Future Web

[NetPL 2017] Toward Building Memory-safe Network Functions with Modest Performance Overhead

Keunhong Lee, Shinae Woo, Sanghyeon Seo, Jihyeok Park, Sukyoung Ryu, and Sue Moon

Proceedings of the 3rd SIGCOMM Workshop on Networking and Programming Languages

[SOAP 2017] Revisiting Recency Abstraction for JavaScript: Towards an Intuitive, Compositional, and Efficient Heap Abstraction

Jihyeok Park, Xavier Rival and Sukyoung Ryu

Proceedings of the International Workshop on the State Of the Art in Java Program Analysis

Best Paper Award

[ICSE 2017 Demo] Analysis of JavaScript Web Applications Using SAFE 2.0

Jihyeok Park, Yeonhee Ryou, Joonyoung Park, and Sukyoung Ryu

Proceedings of the 39th International Conference on Software Engineering Demonstrations track

[Modularity 2014] JavaScript API Misuse Detection by Using TypeScript

Jihyeok Park

Proceedings of the 13th International Conference on Modularity

ACM Student Research Competition 3rd

Honors & Awards

Dec. 2023	고려대학교 우수강좌 , Dept. Computer Science and Engineering, Korea University (COSE212)	Seoul, South Korea
Mar. 2022	Ph.D. Dissertation Award , College of Engineering, KAIST	Daejeon, South Korea
Feb. 2022	An Outstanding Ph.D. Thesis , School of Computing, KAIST	Daejeon, South Korea
May 2021	ACM SIGSOFT Distinguished Paper Award , ICSE 2021	Madrid, Spain
Dec. 2020	PhD Fellowship Award , NAVER Corp.	Daejeon, South Korea
Sep. 2019	Outstanding Teaching Assistant Award , School of Computing, KAIST (CS320)	Daejeon, South Korea
Feb. 2019	Outstanding Teaching Assistant Award , School of Computing, KAIST (CS320)	Daejeon, South Korea
Aug. 2018	Outstanding Teaching Assistant Award , School of Computing, KAIST (CS320)	Daejeon, South Korea
Aug. 2017	Outstanding Teaching Assistant Award , School of Computing, KAIST (CS320)	Daejeon, South Korea
Jun. 2017	Best Paper , International Workshop on the State Of the Art in Java Program Analysis (SOAP)	Barcelona, Spain
Feb. 2017	Outstanding Teaching Assistant Award , School of Computing, KAIST (CS320)	Daejeon, South Korea
Sep. 2016	Outstanding Teaching Assistant Award , School of Computing, KAIST (CS109)	Daejeon, South Korea
Feb. 2016	Magna Cum Laude (GPA: 3.88/4.3) , School of Computing, KAIST	Daejeon, South Korea
Sep. 2014	Workshop 3rd Award , URP Program, KAIST	Daejeon, South Korea
Apr. 2014	ACM Student Research Competition 3rd , International Conference on Modularity 2014	Lugano, Swiss
Mar. 2014	Honor Program , School of Computing, KAIST	Daejeon, South Korea
Jul. 2013	KAIST Presidential Fellowship , School of Computing, KAIST	Daejeon, South Korea

Teaching

TEACHER

2025 Fall	COSE212: Programming Languages , Korea University	Seoul, South Korea
2025 Spring	COSE215: Theory of Computation , Korea University	Seoul, South Korea
2024 Fall	COSE212: Programming Languages , Korea University	Seoul, South Korea
2024 Spring	COSE215: Theory of Computation , Korea University	Seoul, South Korea
2024 Spring	AAA705: Software Testing and Quality Assurance , Korea University	Seoul, South Korea
2024 Spring	SWS121: Secure Programming , Korea University	Seoul, South Korea
2023 Fall	COSE212: Programming Languages , Korea University	Seoul, South Korea
2023 Spring	COSE215: Theory of Computation , Korea University	Seoul, South Korea

TEACHING ASSISTANT

2019 Fall	CS320: Programming Languages course , KAIST	Daejeon, South Korea
2019 Spring	CS320: Programming Languages course , KAIST	Daejeon, South Korea
2018 Fall	CS408: Computer Science Project course , KAIST	Daejeon, South Korea
2018 Fall	CS320: Programming Languages course , KAIST	Daejeon, South Korea
2018 Spring	CS320: Programming Languages course , KAIST	Daejeon, South Korea
2017 Fall	CS492: Special Topics in Computer Science <Program Analysis> , KAIST	Daejeon, South Korea
2017 Fall	CS320: Programming Languages course , KAIST	Daejeon, South Korea
2017 Spring	CS320: Programming Languages course , KAIST	Daejeon, South Korea
2016 Fall	CS320: Programming Languages course , KAIST	Daejeon, South Korea
2016 Spring	CS109: Programming Practice course , KAIST	Daejeon, South Korea

Talks

Aug. 2025	SIGPL Summer School 2025 , 안전한 자바스크립트 언어 생태계 확보 기술	Seoul, South Korea
Jul. 2025	Programming Language Standardization and Specification , Trusted JavaScript Language Environments with ESMeta	Bergen, Norway
Jun. 2025	Real-World Programming Language Specification (RPLS) , Trusted JavaScript Language Environments with ESMeta	Seoul, South Korea
May 2025	고려대학교 CS 콜로퀴움 , 기계화 명세를 이용한 자바스크립트 언어의 설계와 구현	Seoul, South Korea
Apr. 2025	Software Convergence Symposium 2025 (SWCS) , 기계화 명세를 통한 안전한 자바스크립트 언어 생태계 확보 기술	Seoul, South Korea

Nov. 2024	Ulsan National Institute of Science and Technology (UNIST) , Language Design and Implementation using JavaScript Mechanized Specification	Ulsan, South Korea
Jun. 2024	Instituto Superior Técnico of University of Lisbon , Language Design and Implementation using JavaScript Mechanized Specification	Lisbon, Portugal
May 2024	한국과학영재학교 수리정보과학부 콜로퀴움 , JavaScript 언어 생태계 자동화 연구를 하기까지의 여정	Busan, South Korea
Nov. 2023	고려대학교 융합보안대학원 세미나 , 자바스크립트 엔진 보안과 퍼징	Seoul, South Korea
Oct. 2023	고려대학교 CS 콜로퀴움 , 기계화 명세를 이용한 자바스크립트 언어의 설계와 구현	Seoul, South Korea
Aug. 2023	SIGPL Summer School 2023 , PL 구현체를 위한 새로운 커버리지를 제안하기까지의 여정	Chuncheon, South Korea
May 2023	고려대학교 고려대학교 KUGODS 연사 초청 특강 , 프로그래밍 언어 연구란?	Seoul, South Korea
Apr. 2023	고려대학교 KU-KIST 융합대학원 저널클럽 , Guiding the Design and Implementation of JavaScript using Mechanized Specification	Seoul, South Korea
Mar. 2023	고려대학교 CS 콜로퀴움 , 기계화 명세를 이용한 자바스크립트 언어의 설계와 구현	Seoul, South Korea
Mar. 2023	고려대학교 소프트웨어 분석 연구실 세미나 , 기계화 명세를 이용한 JavaScript 언어의 설계와 구현	Seoul, South Korea
Jan. 2023	KCSE 2023 , Automatically Deriving JavaScript Static Analyzers from Specifications using Meta-Level Static Analysis	Pyeongchang, South Korea
Dec. 2022	KSC 2022 , Automatically Deriving JavaScript Static Analyzers from Specifications using Meta-Level Static Analysis	Jeju, South Korea
Feb. 2022	SIGPL Winter School 2022 , JavaScript Static Analysis for Evolving Language Specifications	Online
Feb. 2022	STAAR Workshop , JavaScript Static Analysis for Evolving Language Specifications	Online
Jan. 2022	The 88th meeting of Ecma TC39 , JavaScript Static Analysis for Evolving Language Specifications	Online
Jan. 2022	KCSE 2022 , JSTAR: JavaScript Specification Type Analyzer using Refinement	Online
Jan. 2022	KCSE 2022 , JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification	Online
Jan. 2022	Agoric , JavaScript Static Analysis for Evolving Language Specifications	Online
Dec. 2021	KSC 2021 , JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification	Online
Dec. 2021	Département d'Informatique de l'École Normale Supérieure , JavaScript Static Analysis for Evolving Language Specifications	Online
Oct. 2021	EIRIC , JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification	Online
Sep. 2021	KAISTPL Workshop 2021 , JavaScript Static Analysis for Evolving Language Specifications	Online
Jun. 2021	KAIST Prosys Lab , Towards Co-evolution of JavaScript Specification and Tools	Daejeon, South Korea
Jun. 2021	KCC 2021 , JISET: JavaScript IR-based Semantics Extraction Toolchain	Jeju, South Korea
Nov. 2019	National University of Singapore , Update-Tolerant JavaScript Static Analysis for Frequently Released ECMAScript	Singapore

Software

ESMeta: ECMAScript Specification (ECMA-262) Metalanguage

PROGRAMMING LANGUAGE RESEARCH GROUP, KAIST

Jan. 2022 - PRESENT

- Main developer of ESMeta
- Developed in Scala
- URL: <https://github.com/es-meta/esmeta>

SAFE: Scalable Analysis Framework for ECMAScript

PROGRAMMING LANGUAGE RESEARCH GROUP, KAIST

Mar. 2016 - PRESENT

- Main developer of SAFE version 2.0
- Developed in Scala
- URL: <https://github.com/sukyoung/safe>

Activities

COMMITTEE MEMBERS

Jul. 2027	FSE 2027 , Program Committee (PC) Member	Shenzhen, China
Nov. 2026	CCS 2026 , Program Committee (PC) Member	Hague, Netherlands
Oct. 2026	ASE 2026 , Program Committee (PC) Member	Munich, Germany
Sep. 2026	ICSME 2026 , Program Committee (PC) Member	Benevento, Italy
Jun. 2026	SOAP 2026 , Program Committee (PC) Member	Boulder, United States

Jun. 2026	PLDI 2026 , Review Committee (RC) Member	<i>Boulder, United States</i>
Mar. 2026	SAC 2026 (PL Track) , Program Committee (PC) Member	<i>Thessaloniki, Greece</i>
Oct. 2025	OOPSLA 2025 , Review Committee (RC) Member	<i>Singapore</i>
Oct. 2025	SAS 2025 , Program Committee (PC) Member	<i>Singapore</i>
Sep. 2025	SCAM 2025 , Program Committee (PC) Member	<i>Auckland, New Zealand</i>
Jun. 2025	RPLS 2025 , Organizing Committee (PC) Member	<i>Seoul, South Korea</i>
Apr. 2025	SAC 2025 (PL Track) , Program Committee (PC) Member	<i>Sicily, Italy</i>
Oct. 2024	APLAS 2024 Student Research Competition & Posters , Selection Committee (SC) Member	<i>Kyoto, Japan</i>
Oct. 2024	SPLASH 2024 Doctoral Symposium , Program Committee (PC) Member	<i>Pasadena, United States</i>
Apr. 2024	SAC 2024 (PL Track) , Program Committee (PC) Member	<i>Avila, Spain</i>
Apr. 2024	ICSE 2024 , Program Committee (PC) Member	<i>Lisbon, Portugal</i>
Dec. 2023	OOPSLA 2023 , Extended Review Committee (ERC) Member	<i>Lisbon, Portugal</i>
Nov. 2023	APLAS 2023 , Program Committee (PC) Member	<i>Taipei, Taiwan</i>
Dec. 2022	OOPSLA 2022 , Extended Review Committee (ERC) Member	<i>Auckland, New Zealand</i>
Dec. 2022	APLAS 2022 , Program Committee (PC) Member	<i>Auckland, New Zealand</i>
Jul. 2022	ISSTA 2022 , Web Co-Chair	<i>Daejeon, South Korea</i>

ARTIFACT EVALUATION COMMITTEE (AEC) MEMBERS

Dec. 2023	OOPSLA 2023 , Artifact Evaluation Committee (AEC) Member	<i>Lisbon, Portugal</i>
Dec. 2022	OOPSLA 2022 , Artifact Evaluation Committee (AEC) Member	<i>Auckland, New Zealand</i>
Oct. 2019	POPL 2019 , Artifact Evaluation Committee (AEC) Member	<i>Louisiana, United States</i>
Jul. 2019	OOPSLA 2019 , Artifact Evaluation Committee (AEC) Member	<i>Athens, Greece</i>
Jul. 2018	OOPSLA 2018 , Artifact Evaluation Committee (AEC) Member	<i>Massachusetts, United States</i>
Jun. 2018	SAS 2018 , Artifact Evaluation Committee (AEC) Member	<i>Freiburg im Breisgau, Germany</i>
May. 2018	ISSTA 2018 , Artifact Evaluation Committee (AEC) Member	<i>Amsterdam, Netherlands</i>

JOURNAL REVIEWERS

2026	Transactions on Software Engineering and Methodology (TOSEM) , Reviewer
2025	Transactions on Software Engineering and Methodology (TOSEM) , Reviewer
2024	IEEE Transactions on Software Engineering (TSE) , Reviewer
2023	IEEE Transactions on Software Engineering (TSE) , Reviewer
2022	IEEE Transactions on Software Engineering (TSE) , Reviewer

OTHERS

Sep. 2015 - Feb. 2019	Samsung Electronics , Reviewer for C++ Code Reviewing Exams	<i>Daejeon, South Korea</i>
Jul. 2018	École normale supérieure , Internship Program (Professor: Xavier Rival)	<i>Paris, France</i>
Sep. 2017	NII Shonan Meeting , Topic: Memory Abstraction	<i>Shonan, Japan</i>
Feb. 2015 - Jun. 2015	INSA de Toulouse , Exchange Student Program	<i>Toulouse, France</i>
Jun. 2014 - Nov. 2014	KOFAC , Undergraduate Research Program (URP) for Fusion of Creatives	<i>Daejeon, South Korea</i>
Dec. 2013 - Jun. 2014	School of Computing, KAIST , Undergraduate Research Project (URP) Program	<i>Daejeon, South Korea</i>